Utilizing Deep Learning Models to Develop a Human Behavior Recognition System for Vision-Based School Violence Detection

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***Abstract*— School violence is a worrying problem in every school around the world and is becoming more and more complicated with many unfortunate incidents occurring on school campuses. This study aims to validate the effectiveness of the system for detecting violent acts in school environments. The dataset for training includes 48 videos with full HD resolution (1080x1920), frame rate 30 fps and different lengths simulating violence that we made and 11 videos about bully activities and normal activities from the internet. Besides that, the dataset for testing includes 3 videos we made and 7 video from the Internet. The system detects school violence by classifying the behavior of everyone appearing in the video as bullies, victims, and outsiders. The YOWOv2 model [1], including two branches, 3D-CNN using ResNext101, 2D-CNN using Backbone of YOLOv7 and channel fusion module and attention mechanism (CFAM) block, is transfer learned to solve the problem. When validating, the best model achieved a mean average precision (mAP) of 75.7355%. The AP for the "bully" class is 77.1823%, for the "victim" class is 78.22004%, for the "outsider" class is 71.8043%. When testing, the best model achieved 65.2007% for mAP, 63.337% AP for "bully" class, 78.93005% AP for "outsider" and 53.3352% AP for "victim" class. This model can be improved and developed into an automatic assessment method to detect violent acts in the school environment in real-time to meet actual monitoring needs.**

Keywords*— School violence, Human action detection, one-stage detection, Spatial-temporal action detection)

I. INTRODUCTION

This study emphasizes the application of Deep Learning (DL) - Machine Learning (ML) to identify negative human behavior in videos. Specifically, the research aims to solve the increasing problem of violent behaviors of students in the school environment in today's modern society. This system will be deployed at strategic locations in schools to collect and classify data about student behaviors. When violent behavior is detected, the system will automatically notify teachers or security guards to promptly prevent it, contributing to protecting security in the educational environment.

The application of DL – ML in school violence detection problems can help computer systems, through cameras, identify violence actions in school areas automatically, quickly, accurately, and. more effective, while helping to improve student safety and protection. These systems are currently being researched and developed worldwide.

In a detailed review article [2], numerous research studies conducted between 2011 and 2021 were covered, and they were divided into two main categories based on whether the training data were obtained from surveillance footage or not. Most of the methods prefer the DL-ML approach for better feature attraction. Some of the most commonly used DL architectures recently mentioned include RNN, CNN, and combinations of multiple architectures such as RNN and CNN [3], CNN and LSTM [4], and 2D and 3D CNN [5]. However, it will be extremely difficult to find studies specifically focused on detecting violent behavior in school environments among the papers mentioned in [2]. And most of the above studies were trained and evaluated based on available data sets such as Hockey Fight, UCF-Crime, movies with violent scenes, so it can be seen that the data related to violence in educational environments is also lacking greatly as these actions are unexpected in school areas.

An early study focused on extracting skeleton data for specific violent actions (VAs) recognition on campus [6]. An ensemble of Long Short-Term Memory (LSTM) networks is used, in which each LSTM network is trained to predict a specific VA, to lower the risk of misclassification. However, the limitation of the dataset used is that it focuses on a few actions, such as punching, kicking, and pushing, while school violence can manifest in many other forms.

A different approach for campus violence detection is proposed in [7]. The author uses a C3D (Convolutional 3D) neural network for video-based feature attraction and classification on a custom dataset. This dataset was gathered by mimicking campus violence and normal activities performed by volunteers, achieving a result of 92% accuracy. Additionally, an audio-based model is introduced to indicate whether the audio corresponds to violence or non-violence. Three databases were used, achieving accuracies of 91.67%, 95%, and 88.33% for the CASIA, Finnish, and Chinese databases, respectively. Ultimately, the two results from the

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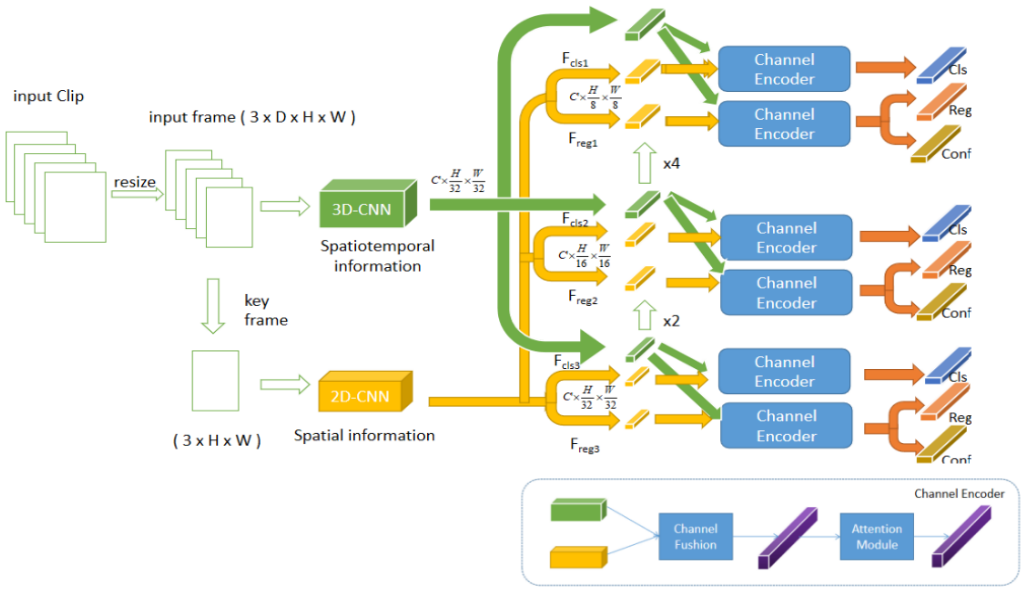


Fig. 1. YOWOv2 model structure reconstructed from article [1]

video-based and audio-based procedures were combined, and a fusion algorithm was applied, resulting in a significant improvement in the average recognition accuracy to 97%.

A recent study presented the School Violence Detection system (SVD) trained on a newly proposed dataset called the Daily School Break dataset (DSB) [8]. The DSB was introduced because the author indicated that current publicly available datasets contained actions and scenes that did not represent real school violence scenarios. The SVD system is based on 2DCNN, and three ImageNet pre-trained models were used for training and evaluation. Eventually, VGG19 was chosen as the best-performing model when trained on the extended dataset, which involved enriching the original dataset with new images downloaded from the internet. The model achieved a result of 95% accuracy.

In a similar approach to creating a new dataset for recognizing school violence, a study called VSiSUG [9] achieved an accuracy of 88% using a proposed CNN-RNN model. The VSiSUG dataset consists of 7240 videos with 18 categories of school violence.

Most of the related works mentioned above have primarily focused on classifying whether a scene is violence or non-violence, or identifying the type of violence occurring [9]. However, this paper proposes a new approach for detecting violent actions in a school environment. The proposed method aims to detect each individual present in the scene and classify their actions as bullying, victim, or outsider. This approach takes into account the specific roles and interactions of individuals within the context of school violence, providing a more comprehensive understanding of the dynamics involved.

II. Dataset

The data collection and preparation process is an important step to ensure the quality and accuracy of the human behavior recognition system. Today, the importance of data is not only a technical issue but also a decisive factor in the success of DL projects. In particular, within the scope of this problem of identifying violent behavior in the school environment, the data preparation process receives the most attention because this is a decisive stage that has a huge impact on the quality and effectiveness of the system accuracy.

Because the goal of this research focuses on detecting violent behavior in the school environment, the dataset needs to meet the following criteria: (1) Diversity of situations in which school violence occurs; (2) The target audience is students and diverse school contexts; (3) Lighting conditions and shooting angles are similar to security cameras; (4) Actual data and social reflection.

The dataset includes 69 video segments with different lengths and a minimum of 1 minute. Among them, 18 videos collected from the internet and 51 self-made videos simulating school violence situations that can appear from many angles and are more than 1 minute long. To ensure data video quality, the recording equipment used has Full HD 1080x1920 resolution and a stable frame rate of 30 FPS. High resolution helps detail violent acts, and stable frame rates determine the system's performance during training. The videos mentioned above were recorded at Ho Chi Minh City University of Technology - National University.

All the videos are divided into frames and each video frame is manually labeled on Roboflow with the categories: bully, victim, and outsider and is assigned individual ids throughout the video. To assess the influence of video length on model performance, each video is segmented into smaller videos of equal duration. Two consecutive cut videos in the dataset are overlapped by half their length. This methodology produces three datasets comprising videos with durations of 32 seconds. Because the action per second doesn't make much difference, only 2 frames are used per second (the 15th and 30th frames). At the same time, each video in the datasets is flipped horizontally to enhance the training data. Then, the datasets are reorganized in the same way as the AVA dataset [10].

From the 69 videos we collected, 59 videos were used for training and 10 videos were used for testing. The training videos were split into a total of 375 sub-videos. Out of these, 350 sub-videos were used for training, and 25 sub-videos were allocated for validation. However, the 10 videos used for testing were not split into sub-videos.

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III. Methodology

YOWOv2 is a single-stage deep learning model that is highly effective in detecting spatiotemporal actions in video streams in real-time. The model has shown superiority in prediction speed, the combination of both spatial and temporal information during training and real-time calculation compared to other serial information processing models, specifically in processing video information. We use the same loss function as proposed in the paper [1], as follows:

$$L\left(\{a_{x,y}\},\{b_{x,y}\},\{c_{x,y}\}\right) = \frac{1}{N_{pos}}\sum_{x,y} L_{conf}\left(\hat{c}_{x,y}, c_{x,y}\right) + \frac{1}{N_{pos}}\sum_{x,y} I_{\{\hat{a}_{x,y}>0\}} L_{cls}\left(\hat{a}_{x,y}, a_{x,y}\right) + \frac{\lambda}{N_{pos}}\sum_{x,y} I_{\{\hat{a}_{x,y}>0\}} L_{reg}\left(\hat{b}_{x,y}, b_{x,y}\right) \quad (1)$$

where L_{conf}, L_{cls} and L_{reg} are the confidence loss, class loss and region loss, respectively. $a_{x,y}$, $b_{x,y}$, and $c_{x,y}$ represent predictions for classification, regression, and confidence, respectively, while $\hat{a}_{x,y}$, $\hat{b}_{x,y}$, and $\hat{c}_{x,y}$ correspond to the ground truths for these predictions. N_{pos} represents the positive sample, and λ is the loss balance factor, which is assigned a value of 5 in this implementation. $I_{\{\hat{a}x,y>0\}}$ equals 1 if $\hat{a}_{x,y} > 0$ and 0 otherwise.

As shown in **Fig. 1**, the YOWOv2 model includes a 2D-CNN branch to extract spatial features at the current image frame (key frame) and a 3D-CNN branch used to extract features between space and time from the current frame and previous frames. Finally, a channel fusion and attention mechanism (CFAM) block is used to combine the information from the above two branches and create a combined feature used to determine the action for each frame.

Research [1] also shows that the model has the best results on different datasets when the 2D-CNN branch uses the Backbone of YOLOv7 [11] and the 3D-CNN branch uses the backbone of ResNext-101 [12]. The model is trained with 5 epochs and the weights are initialized from scratch for the datasets mentioned in II. Before entering the training process, image frames from the data set are scaled to 224 x 224 pixels and there is no additional image processing.

A warning mechanism is added to the model to consider whether or not violent behavior is occurring based on the action prediction results from the model. Specifically, within 8 frames up to the current frame (equivalent to 0.5 second with image collection quality from the camera being 30 FPS), when a frame appears in the "bully" class, that frame will be counted as 1 point. After 8 frames, the score of those 16 frames will be calculated, if the score is equal 4 or more, the current frame will display a warning (Alert: Bullying), otherwise there will be no warning (Alert: Normal). This ensures that even if sometimes the frame does not appear the "bully" class, but the warning will still be maintained if most the predicted information part of the model is bullying.

IV. Implementation

When training data, we changed the code and label map from source of YOWO [13] and YOWOv2 to fit our data (64 frames per sub video train of our dataset compared to 900 frame of AVA dataset and changed number of labels from 80 classes to 3 classes).

Moreover, we set up 16 frames for each batch and processed 8 batches per iteration. Training the YOWOv2 model on our dataset took 5 hours for 5 epochs using a GPU P40 24GB. In contrast, training the YOWO model with the same dataset took approximately 16 hours for 5 epochs using a free account on Google Colab. During the training process, we used the AdamW optimizer with an initial learning rate of 0.00001, decaying the learning rate by half at the 3rd, 4th, and 5th epochs, respectively.

When testing to calculate AP and mAP, we modified the code to calculate the maximum frame for each video based on the maximum number of frames in each video in the test dataset, instead of using 64 frames each sub-video as in the training process. For testing visualization, we collected 16 continuous frames and fit them to our model. The model extracted features from frames 1 to 16 and then predicted the bounding box and action of each person in the 16th frame. We then removed the 1st frame, added the 17th frame to the list, and repeated the process. The model extracted features from frames 2 to 17 to predict the bounding box and action of each person in the 17th frame, and so on. The alert mechanism started calculating points when the model predicted the first frame, meaning the mechanism started calculating points from the 16th frame. During the initial period, the alert was set to "normal".

V. Results And Discussions

After the training progress, the results for each epoch are shown in **Fig. 2**. Although the mAP result is not too high (the highest is 75.7355%), this result can be further improved when a new dataset with a longer duration is used to adjust parameters such as batch and epoch, this result can be further improved when a new dataset with a longer duration is used to adjust parameters such as batch and epoch.

Based on **Fig. 2**, its shows that the AP of class "outsider"

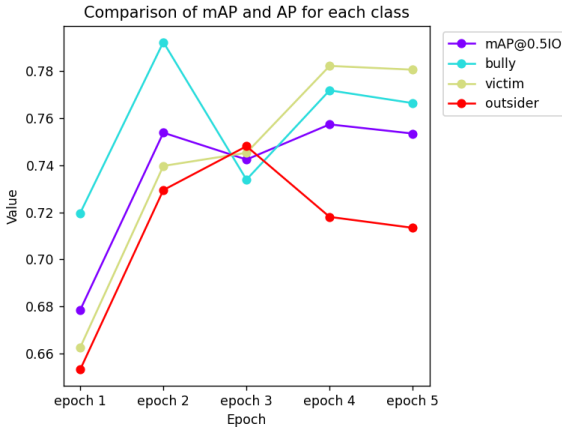


Fig. 2. Visualization for each epoch

is smaller than AP of class "bully" and "victim", it can be explained by the dataset has lot of scenes people walking near together, make a crowed and it can be confused to "bully" and "victim" person.

Identifying distant objects is also sometimes hindered because the input frame is reduced in size to only 224 x 224 pixels, causing data loss and making the model unable to clearly identify objects. Leading to a model cannot detect a

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Fig. 3. Results from test video with red, blue and green bounding box stand for “bully”, “victim”, and “outsider” class, respectively

person clearly (**Fig. 3d**, **e**, **f**) or confuse the action of people (**Fig. 3b**), as the person was grabbing the head of the victim, but this action was identified as belonging to the “outsider” class. The reason is that the model mistakenly thought that the bully did not have a direct impact on the subject being bullied and was unclear, causing the model to confuse the “outsider”

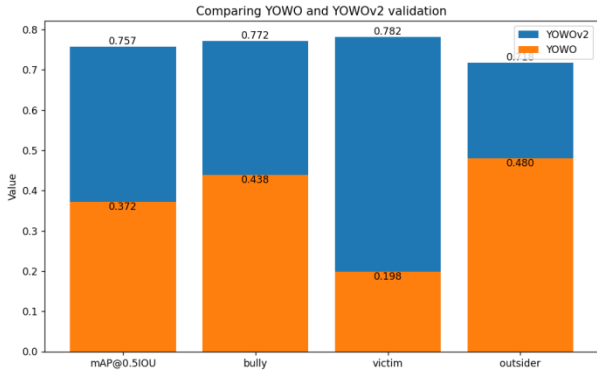


Fig. 4. Comparing YOWO and YOWOv2 in validation dataset

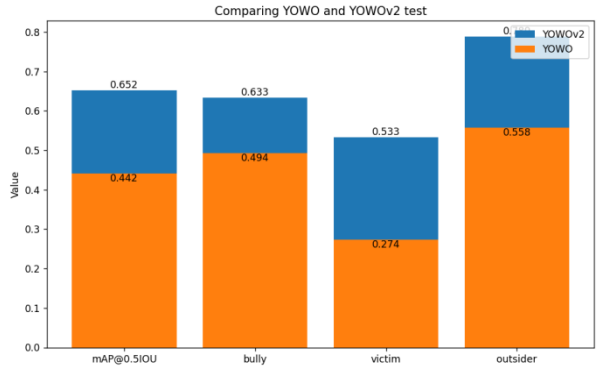


Fig. 5. Comparing YOWO and YOWOv2 in test dataset

with normal gestures. However, when bully gestures were more obvious, such as grabbing clothes, hand or pushing (such as **Fig. 3a** and **Fig. 3c**), the model can distinguish between the “bully” class and the “outsider” class.

As **Fig. 3e**, the test results show that students sometimes engage in joking activities, causing the system to have a negative warning about bullying including 2 classes “bully” and “victim”. But in general, the system has not yet turned-on bullying warnings due to the frame-by-frame scoring mechanism mentioned in III. But these warnings also help us observe dangerous pranks of students, they can push or pull and make someone fall causing injury. After a few seconds, the model detects that the student does not engage in bullying, so the model predicts him to belong to an “outsider” class (**Fig. 3f**).

Fig. 3d and **Fig. 3f** are the result of the model operating on a school campus when no one is bullied, the model identifying everyone as belonging to the “outsider” class (in this case it could be understand that these people did not commit any acts of bullying).

In general, during the testing process. The model can still show correct predictions for the majority of the ongoing events captured by the camera. However, the current drawback of that model is that the FPS is quite low, only approximate 6 FPS when reading frames extracted from security camera videos, which makes it difficult to apply the model in real environments that is still not feasible. The reason is because the length of the clip is up to 8 frames. If we reduce the length of the clip, we can increase the FPS but at the cost of decreasing the accuracy of the model.

Compared with model YOWO, which was trained with the same dataset of this research, we can see that mAP and AP

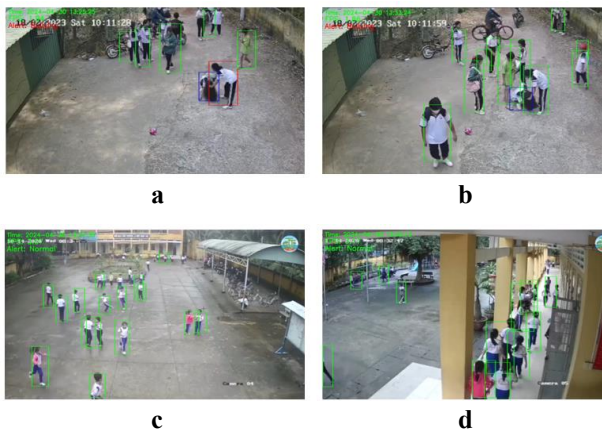


Fig. 6. Result of model YOWO with the same data

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for each class of YOWOv2 is better than YOWO, as are shown in **Fig. 4** and **Fig. 5**.

As **Fig. 4** and **Fig. 5** it can be clearly seen that the AP of each class and mAP results of model YOWO are worse than YOWOv2 when testing, it can be explained by the model 2D architecture of YOWOv2 is the backbone of YOLOv7, is better than Darknet19[14], which is the 2D architecture of YOWO.

And as shown in **Fig. 6**, we can see that the YOWO model does not detect all people in close range, the FPS does not improve much compared to YOWOv2 (also approximately 4 FPS or smaller), besides determining the wrong classes for each object (**Fig. 6b**), and the image is blurred. These results show that YOWOv2's ability to identify small actions is much better than YOWO, FPS and reliability is also higher.

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